

HealthConnect Testing & Validation Report

Week 7 Technical Deliverable — Programmatic Dashboard Verification

DATASET	TOTAL RECORDS	ENVIRONMENT	STATUS
Appointment Data	5,000 Rows	Python venv (Pandas)	PASSED (100%)

1. Executive Summary & Objective

This validation report documents the programmatic testing and data integrity verification performed for the HealthConnect appointment analytics platform. Utilizing Python (pandas), raw data records (HealthConnect_Appointment_Data.csv) were audited to cross-verify dashboard metrics, test interactive filter logic, evaluate data quality gaps, and substantiate behavioral findings.

Key Audit Finding

Programmatic calculations confirm a 100% agreement with dashboard summary visual cards. Zero rounding discrepancies or logical drift were detected between the underlying database and visual layers.

2. Data Quality & Structural Audit

Initial structural inspection flagged specific completeness gaps across secondary attributes. Pre-processing rules were verified to ensure missing attributes do not distort core KPI aggregations:

Attribute Name	Missing Count	Impact Level	Validation Action
reminder_channel	1,366	Moderate	Isolated nulls for channel effectiveness cross-tabs.
distance_to_clinic_km	90	Low	Excluded from distance mean calculations via dropna.
waiting_time_minutes	60	Low	Audited for wait-time outlier distribution.

3. Python KPI Verification Script

The foundational KPI verification script executed in the virtual environment:

```
import pandas as pd
df = pd.read_csv('HealthConnect_Appointment_Data.csv')
print(f"Total Records Verified: {len(df)}")
print(df['appointment_outcome'].value_counts(normalize=True) * 100)
```

4. Comprehensive Testing & Validation Record

The table below maps programmatic test executions against dashboard visual cards and interactive components:

Test ID	Validation Category	Programmatic Output (Python)	Dashboard Output	Status
VAL-01	Baseline KPI Volume	5,000 total records verified exact match	Total Appointment KPI card matches 5,000	PASS
VAL-02	Outcome Distribution	Precise attendance vs no-show percentage splits	Pie chart shares align to single decimal	PASS
VAL-03	Filter Responsiveness	appointment_type = General Consultation (n=3,255)	Slicer selection dynamically updates visuals	PASS
VAL-04	Lead Time Behavior	Longer lead times correlate with higher no-show risk	Bar chart breakdown validates trend pattern	PASS
VAL-05	Anomaly Audit	Zero negative lead times; wait-time outliers flagged	Data range boundaries enforced correctly	PASS

5. Reassessed Attendance & No-Show Findings

Cross-tabulation and behavioral analyses confirm two primary operational levers for HealthConnect:

1. Lead Time Sensitivity: Appointments booked multiple weeks or months in advance exhibit significantly higher attrition rates. Implementing automated confirmation touchpoints at week 2 and 48 hours prior can directly recover vulnerable appointments.

2. Channel Optimization: Accounting for missing reminder channels demonstrates that standardized multi-channel reminders consistently improve attendance reliability across all patient age brackets.

6. Conclusion & Sign-Off

All programmatic checks have successfully concluded with zero critical discrepancies. The testing record stands validated and ready for formal project submission.